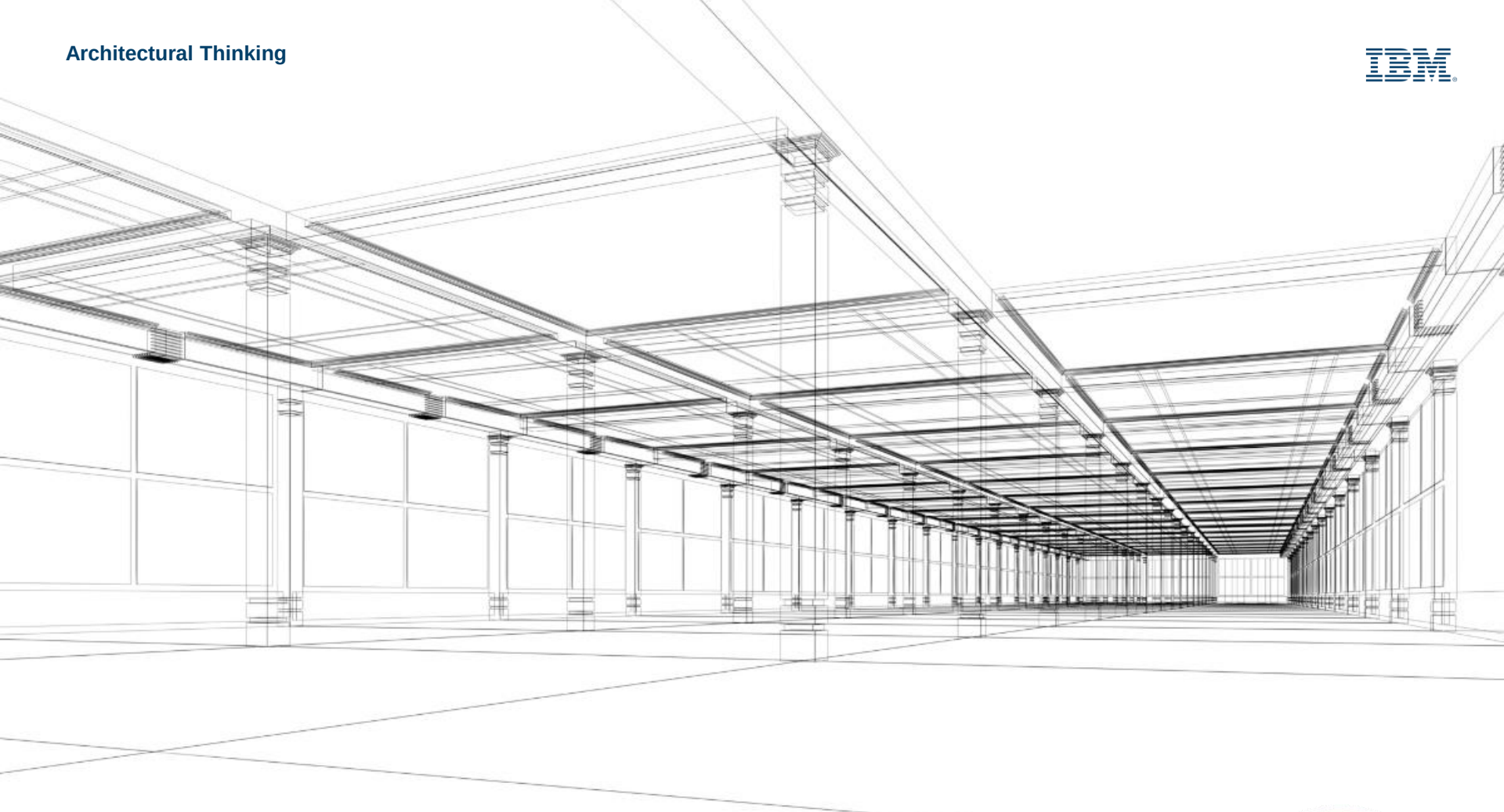




What is Architecture?



Module outline



► Introduction and objectives

What is architecture?

IBM's Architects

Summary and references



At the end of this module, you should be able to:

- Explain what is meant by architecture and how we use it to help us design viable IT systems
- Understand why architecture is important in the modern age of fast-paced change
- Describe what an IT Architect is and what one does in the design on IT systems
- Describe the relationship between:
 - Enterprise architecture and solution architecture
 - Business architecture and IT architecture



Spend five minutes thinking about the following questions:

- What is your definition of architecture?
- What is the difference between architecture and design?
- Why is architecture important to IT systems design, and why do projects need it?
- What is the role of the IBM Architect?
- What diagrams and models do you currently use to represent architectures?
- What do *you* think is meant by architectural thinking?



Here are some popular misconceptions about architecture:

- Architecture is just paper.
- Architecture and design are the same thing.
- Architecture and infrastructure are the same thing.
- <My favorite technology> is the architecture.
- A good architecture is the work of a single architect.
- Architecture is simply structure.
- Architecture can be represented in a single blueprint.
- Architecture cannot be measured or validated.
- Architecture is a science.
- Architecture is an art.



Module outline



Introduction and objectives

► **What is architecture?**

IBM's Architects

Summary and references



Architecture (as an artefact):

*The architecture of a system describes its overall **static structure and dynamic behaviour**. It models the **system's elements** (which for IT systems are software, hardware, and its human users), the **externally manifested properties** of those elements, and the static and dynamic relationships among them.*



Architecture (as a discipline):

*Architecture is an **engineering discipline** that studies methods of designing **IT systems** that provide **a solution to a business problem**. The solution must **satisfy** functional and non-functional **requirements** in a way that best balances **competing stakeholders' concerns** and must take **constraints** into account.*



Think about the following statements and see if you agree with them:

- I don't have time for architecture – it's just a document that no one needs anyway. My solution will work perfectly without the need to produce these documents and/or diagrams
- There are too many architectural artefacts that the method defines. I don't have time to complete all this work even if I think it's got value
- We do not need architecture today. Modern ways of delivery use agile methods that make architecture superfluous
- We use cloud based and "Platform as a Service" environments that "decide" on the architecture and spin out the necessary resources for me behind the scenes
- We use a "Commercial off the Shelf" product (i.e. a package) or "Software as a Service" (SaaS) cloud based solution. It has its own architecture that is hidden from me. Therefore I don't need to develop architecture
- We use Design Thinking workshops to work on the solution. This removes the need to develop architecture



What is the role of architecture? Some of the imperatives for producing architecture:

- **Solution blueprint** that can be reviewed and validated
- **Communication tool** for stakeholders and teams (business, IT delivery and BAU – IT Operations), future reference for maintenance and change, developing neighbouring systems and integration
- **Delivery Environments**: Input into the delivery environments' design and specification, tools and platforms of choice for solution delivery
- **Delivery Planning**: Starting point for developing the project plan (Traditional delivery), solution and feature backlog (Agile delivery), resource requirements and team structure
- **Estimates**: Starting point for cost estimation (labour, licences, platform costs, etc)
- **Enable parallel development**: by understanding how inter-component dependencies can and should be managed
- **Foundation for commercial agreements** (i.e. “This is what you get for the price we have agreed on”)
- **Foundation for DOU** and segregation of responsibilities between various parties – e.g. Client and Solution Provider, Solution Provider and Partner(s)/Subcontractor(s) (e.g. RACI matrices)



Why do we need it?

- Both Functional capabilities and Non-functional requirements & Service Levels may be contractually binding:
 - Failure to achieve targets may result in financial penalties for the IT provider, and/or a loss of business for the Client
 - It is an issue of significant severity if an architecture that meets the specified requirements cannot be established. This may be due to lack of a viable solution in principle (e.g. we cannot solve a given problem with the amount of funding available) or due to poor design and the latter needs to be improved. Regardless, if the answer is negative, the sooner this is known the better for all the stakeholders
- The effort expended should always be proportionate to the risk and return involved. Without a developed architecture, it is rare that all significant risks are known and well understood
- Lack of architecture means that there is no basis for solution cost and delivery time estimates

Architecture can be at any scale, and the scale is defined by the characteristic dimension of the architecture's elements (1 of 4)



Paraphrasing the definition: **Structure** and **behavior** of the **elements** of a system

- What is a system?
- What is an element?



Architecture can be at any scale, and the scale is defined by the characteristic dimension of the architecture's elements (2 of 4)



Paraphrasing the definition: **Structure** and **behavior** of the **elements** of a system

- What is a system?
- What is an element?

It's all a question of **scale**:

- The elements of a hi-fi are its tuner, amplifier, speakers, cables, and remote controls.
- The elements of an amplifier are its case, the power supply, and the circuit board.
- The elements of a circuit board are its resistors, fuses, and integrated circuits.
- The elements of an integrated circuit are transistors and other components.



Architecture can be at any scale, and the scale is defined by the characteristic dimension of the architecture's elements (3 of 4)



Paraphrasing the definition: **Structure** and **behavior** of the **elements** of a system

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- The elements of an integrated circuit are transistors and other components.



Whatever the scale of the system we're focused on, we say:

- 'Above' that scale are the '**requirements**' of the world in which the system sits.
- 'At' that scale is the '**architecture**' between the elements of the system.
- 'Below' that scale is the '**design**' of the insides of the elements of the system.



Architecture can be at any scale, and the scale is defined by the characteristic dimension of the architecture's elements (4 of 4)



Paraphrasing the definition: **Structure** and **behavior** of the **elements** of a system

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- '**Below**' that scale is the '**design**' of the insides of the elements of the system.



So, in our world of IT:

- Our IT system's requirements are modelled as interactions (use cases, and so on.) with actors and other IT systems
- Our IT system's architecture's elements are components and nodes, and so on.
- Our IT system's design or designs model the insides of components, nodes, messages, and connections

Architecture also helps us ensure that our systems are good systems



Architecture is also used to describe best (or preferred) practice, documented as *architecture guidance*, that helps ensure our system is a good system.

Architecture guidance includes:

- Good things
 - Tried and tested parts
 - Tried and tested structures of parts (reference architectures)
- Good ways of doing things
 - Principles and patterns that help ensure the architecture is created in a good way

This architectural catalogue of guidance may be found in:

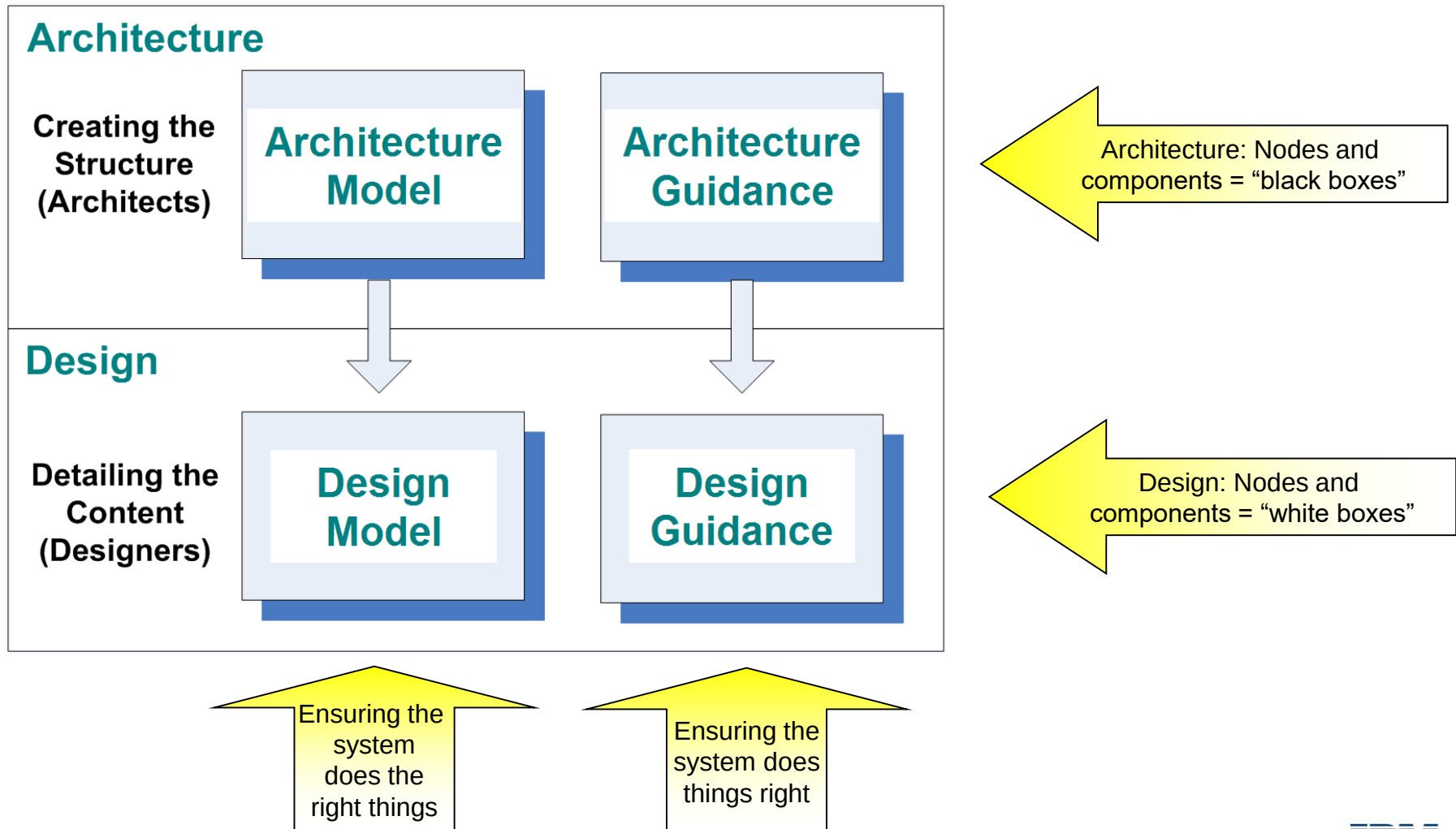
- IBM's own architecture asset databases
- Our clients' enterprise architecture artefacts

See the course video on Architectural Assets available as post-work for an in-detail discussion.

The focus of architecture and design on the target system



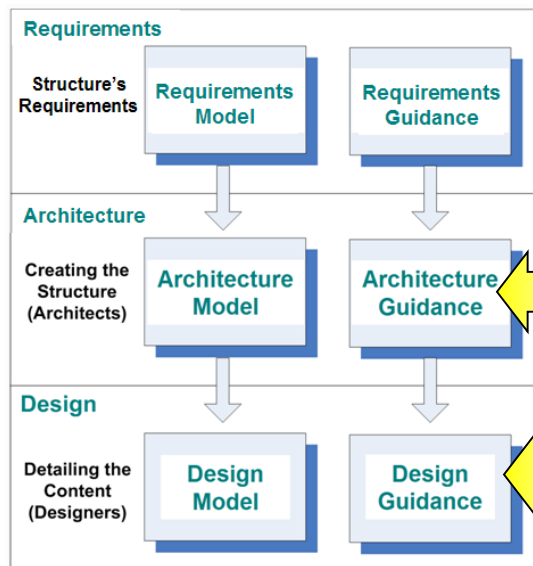
Overall, architecture and design both focus on the content and the context of the target system, ensuring that it does the right things and does things the right way.



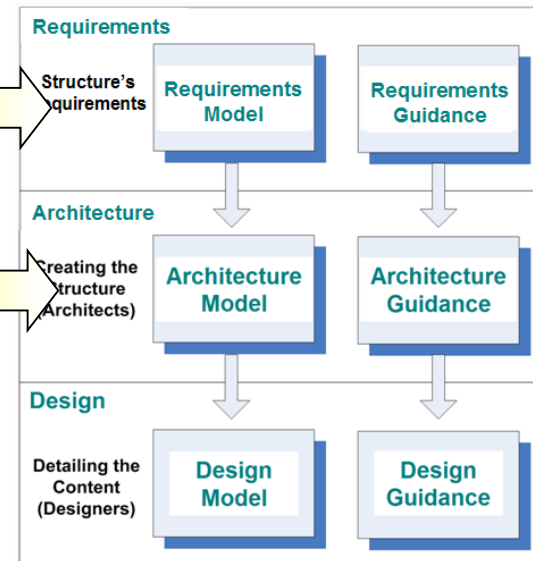
This simple framework can be extended, enabling us to see a repeating pattern across the scales



Architecting the Amplifier



Architecting the Circuit Board





Time for a question



What is the correct definition of architecting?

- A. The process of analyzing and solving, at a high level, technical problems using various business theories
- B. The process of analyzing and solving, at a high level, business problems using various technologies
- C. The process of solution design

What is the value of architecting?



Time for a question



What is the value of architecting?

- A. To deliver the project on time and within budget
- B. To address a business problem with a technical solution involving technologies, people and processes
- C. To bring the vision of the most effective solution into reality

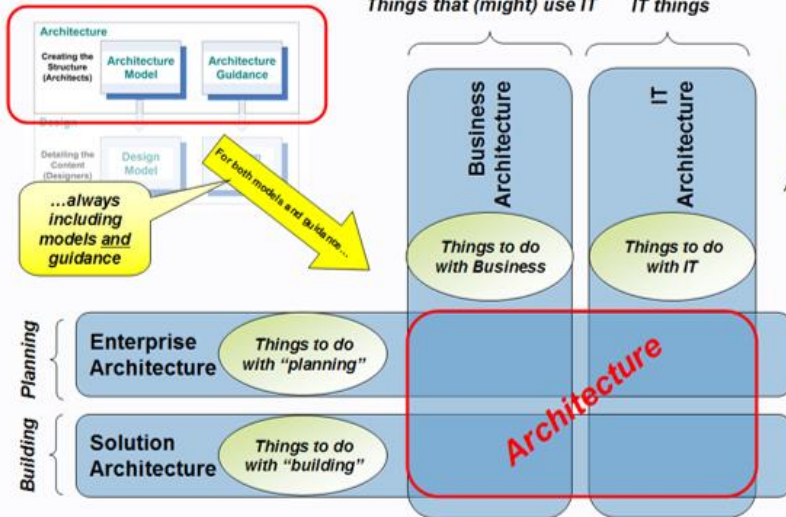


IT architecture is just one of the architectural domains we need to consider if our IT system is to do the right things and do them in the right way.

These multiple architectures can be simplified for each dimension, into a "2 by 2" reference framework



Things that (might) use IT IT things



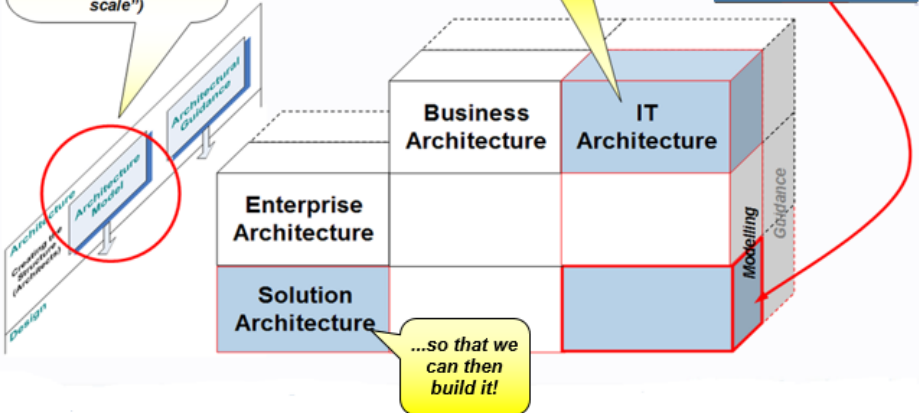
This framework allows us to highlight the focus of this "IT Solution Modeling" class



The AT class addresses the modelling aspect of IT architecture, at solution scale.
(It also briefly surveys "guidance at solution scale")

"We need artifacts that describe the system's IT Architecture..."

For IT Architecture	Systems Models	Guidance for Systems
Enterprise scale or "big" which help plan and organize work	Enterprise Models	Architecture building blocks, design processes, reference models (e.g. PAB, EXO) and other guidance
Project scale or "small" things focused on building something	Programs and Project Models	



Thus this class focuses on the models needed to describe the IT architecture of solution-level systems



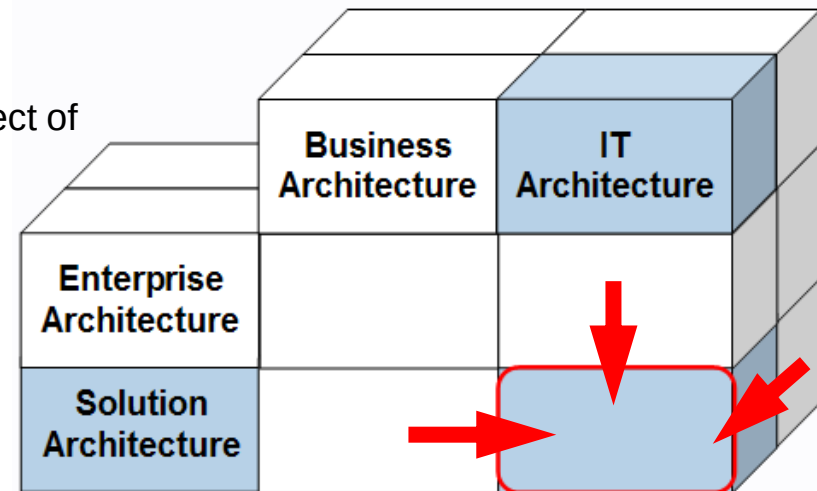
The artifacts covered in this class present models of the target IT system.

They can be **driven by models of the broader business system**, of which this target IT system is a part.

- The target IT system must meet the specific business requirements placed on it.
- Whether it meets those requirements or not is the subject of **Solution Assurance**.

They can be **guided by artifacts** documenting architectural principles, patterns, and building blocks.

- The target IT system must meet the wider architectural requirements placed on it.
- Whether it meets those requirements or not is the subject of **Architectural Conformance**.

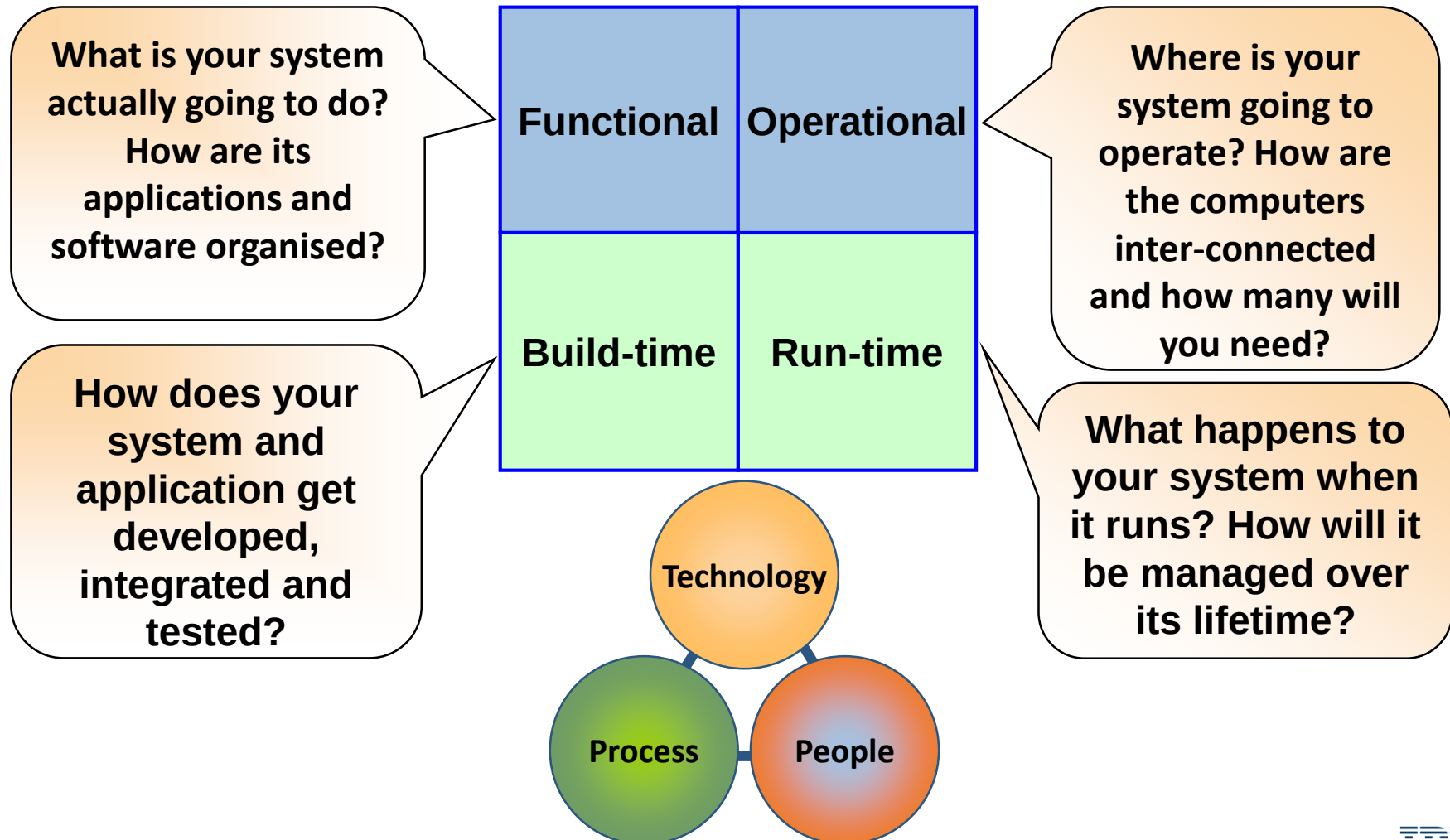


The class focuses on the use of these IT system model artifacts **at the solution scale**, by IT Architects.

- They are also used by Enterprise Architects at the enterprise scale as they plan (in outline) transition initiatives, and so on, in which case they may be **elaborated** by the IT Architect.

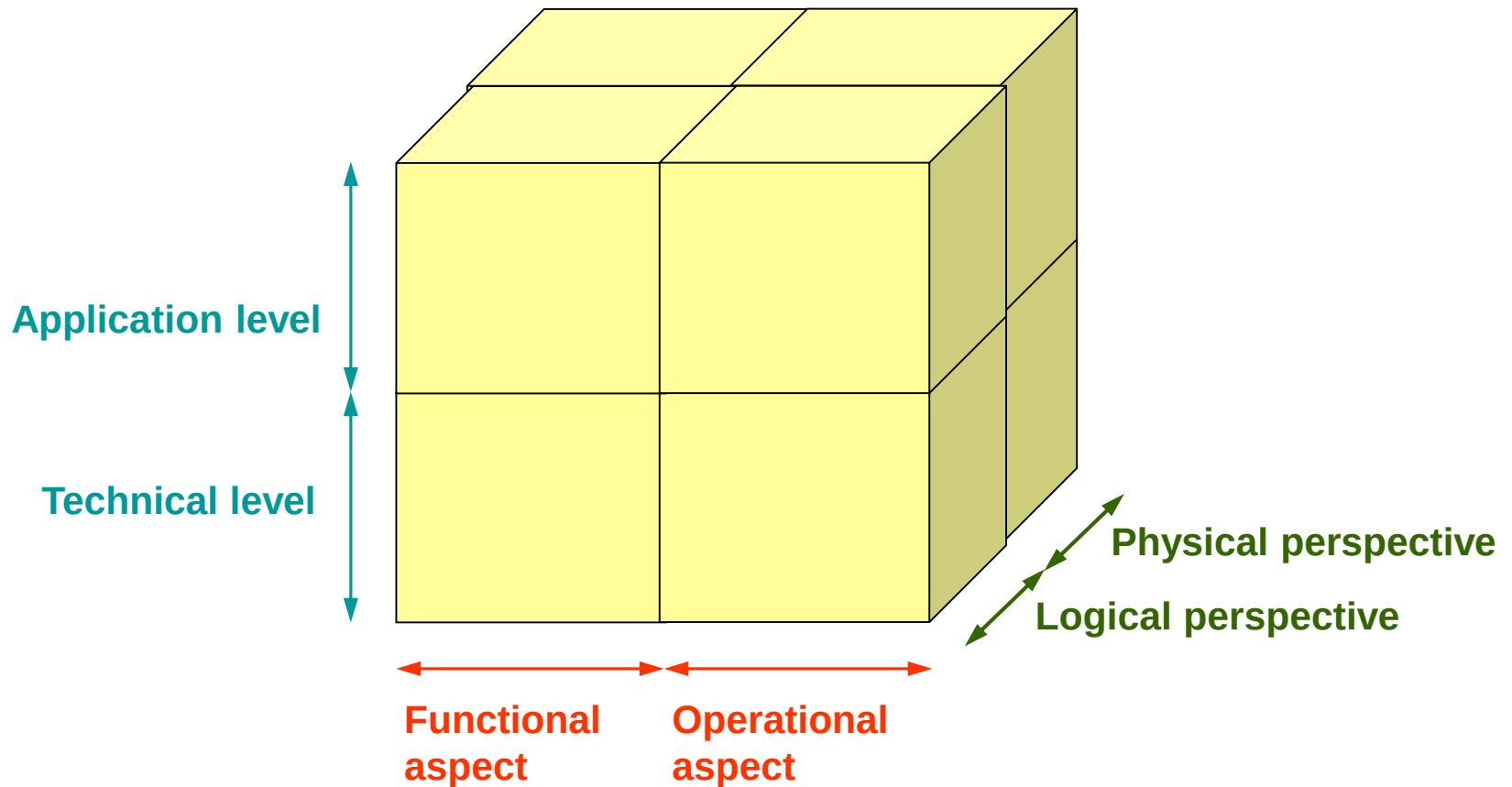


An architect asks questions for the different aspects each with technology, process and people to deliver a complete architecture





Architects look at a problem in a number of dimensions to organise their thoughts



Module outline

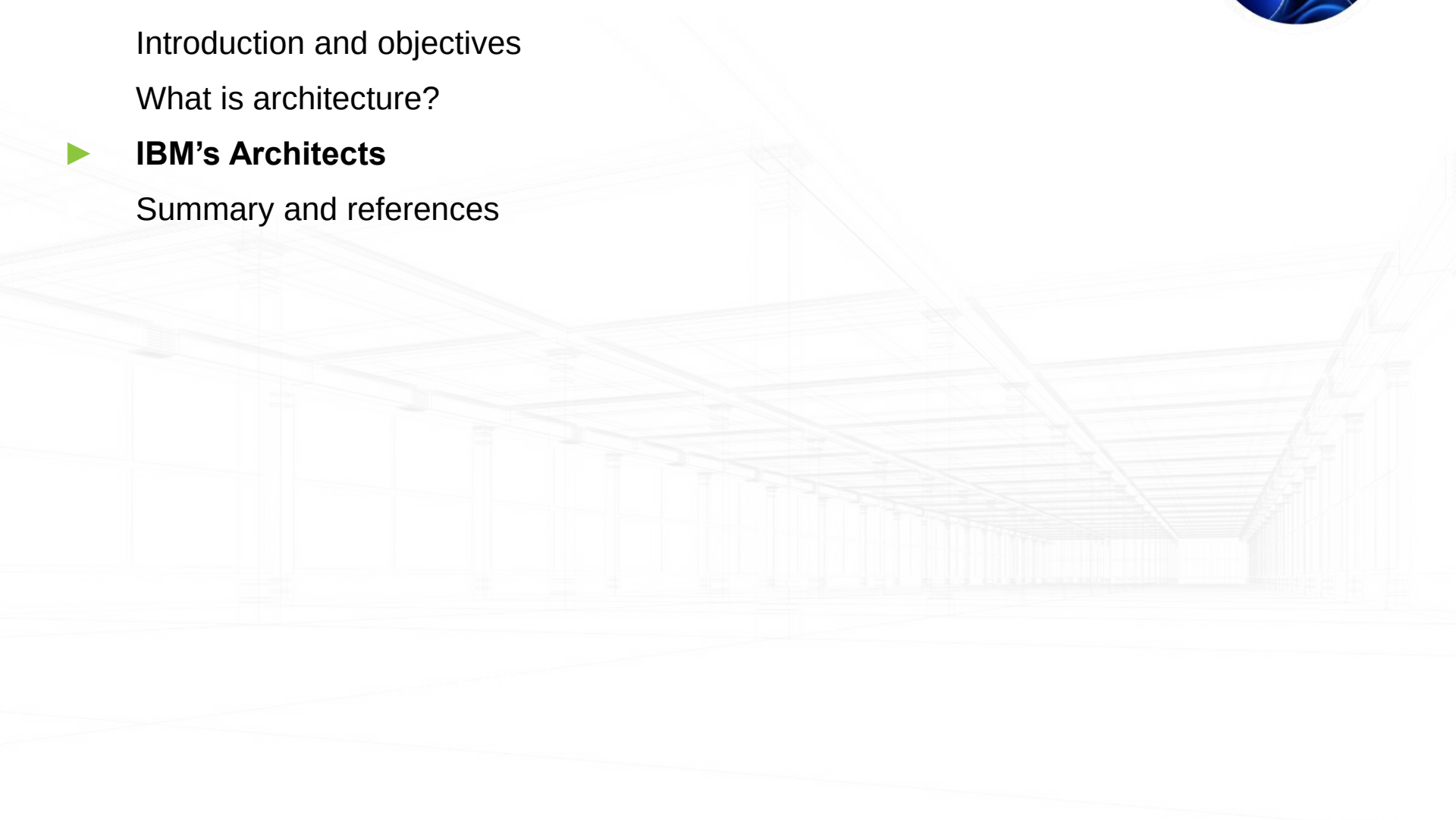


Introduction and objectives

What is architecture?

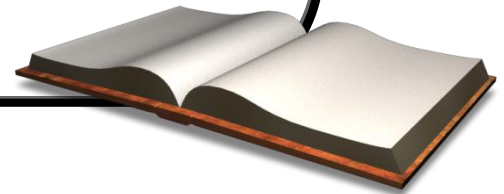
► **IBM's Architects**

Summary and references





Architect: An Architect *defines the structures of solutions* and architectures to address client business problems. The Architect *understands client needs and business constraints*, works in levels of abstraction, applies industry knowledge, and *leverages appropriate business elements and information technology to address those needs*. The solutions are manifested as *business, enterprise, or IT architectures* and can include systems, applications, and process components. They may also involve the application and integration of a broad variety of products, technologies and services, and various architectural styles and domains.





An Architect defines the structures of solutions... [based on] client needs and business constraints... [leveraging] appropriate business elements and information technology to address those needs. ...Solutions [can address] business, enterprise, or IT architecture... domains.

Architects work:

- At the front:
 - Listening to clients
 - Understanding business requirements
 - Negotiating feasible requirements and scope
 - Forming detailed definitions of solution structures
- At the back:
 - During construction
 - Serving as a customer advocate
 - Guiding the team

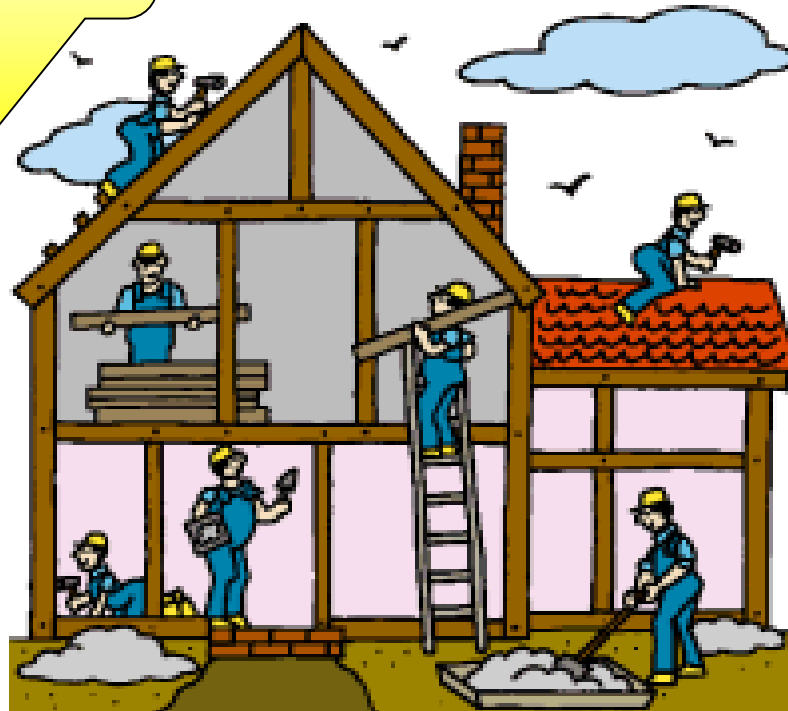
Who is needed to create a new house?



Architect

Many specialists are required to construct the house: Framers, Plumbers, Electricians, and Painters.

Builder, Subcontractors



Occupants

Occupants will live in the house and have wants and needs.

The Architect draws up the plan to meet the occupants' needs and ensures that the builders meet the specifications, understands the influence of building codes, other regulations, legacy, power and gas services, and the neighborhood architecture review committee guidelines.



Building Inspector

The Inspectors ensure that the house is built as per quality standards such as local building codes.

The Project Manager manages the construction of the house and coordinates subcontractors.

Project Manager



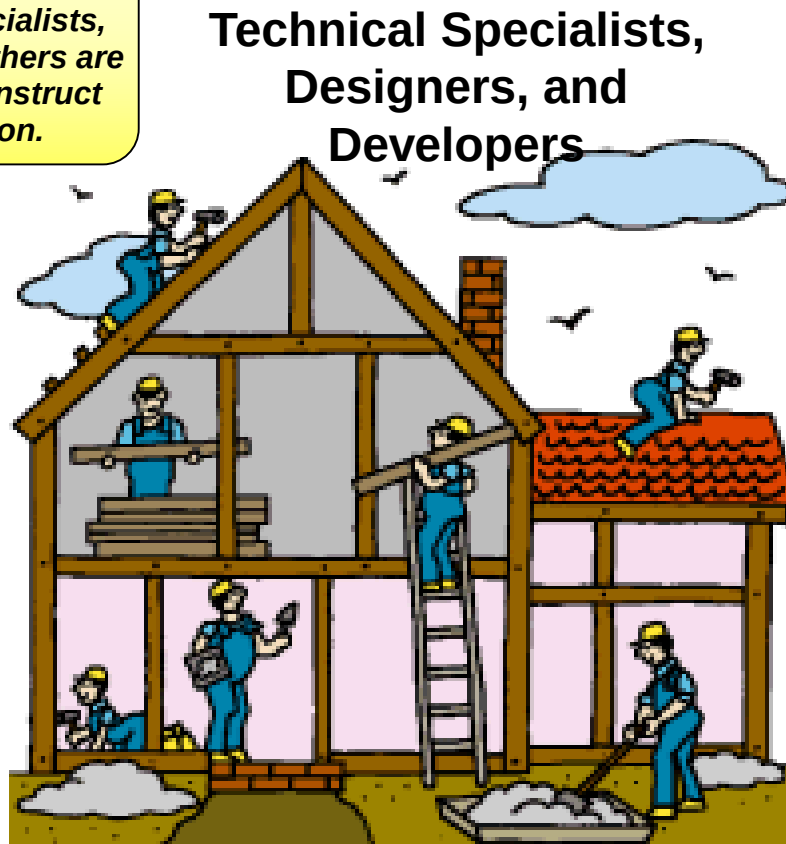
What's this got to do with architecture at IBM?



Solution Architect

Designers, Coders, Package Specialists, Testers, and others are needed to construct the solution.

The Architect creates the solution architecture to meet requirements and ensures that the development meets the specifications, regulations, and passes governance checkpoints.



Technical Specialists, Designers, and Developers



Business Sponsor, Users

Users and their delegates have their requirements.



Enterprise Architect

The Enterprise Architect ensures that the solution architect complies with enterprise standards, governance, and roadmaps.

The Project Manager manages the Solution Delivery.

Project Manager



Comparing Architects and Technical Specialists



Architects

IT Specialists

Architects:

- Define solutions to client business problems. Resultant architectures can include systems, applications, and process components.
- Focus on the front end of the solution life cycle. They may be involved in the construction of the solution as a client advocate.

Technical Specialists:

- Develop proof of concept, design, develop, build, test, and implement systems.
- Are hands-on professionals with in-depth understanding of products, offerings, and services within their specialty.

Analysts

Designers

Testers

Deployment Specialists

Product Specialists

Data Base / Middleware

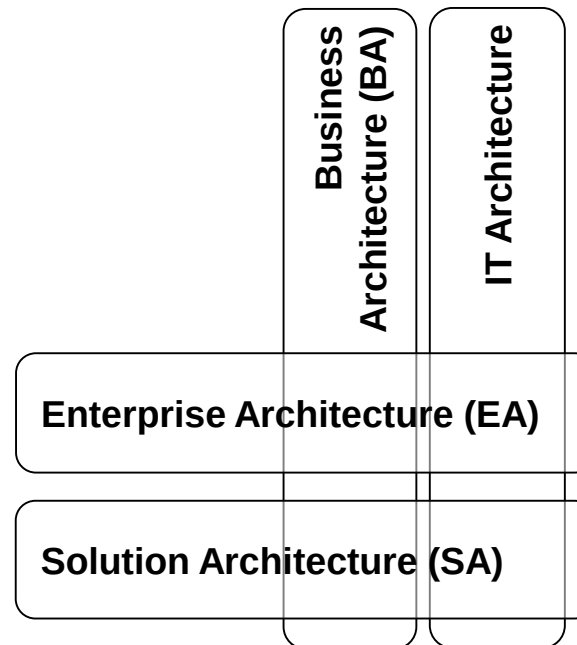
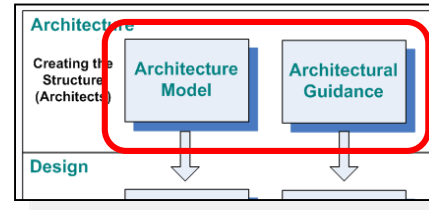
Infrastructure Specialists

Network

Architect supports two distinct dimensions (1 of 5)



Embracing both models and guidance:

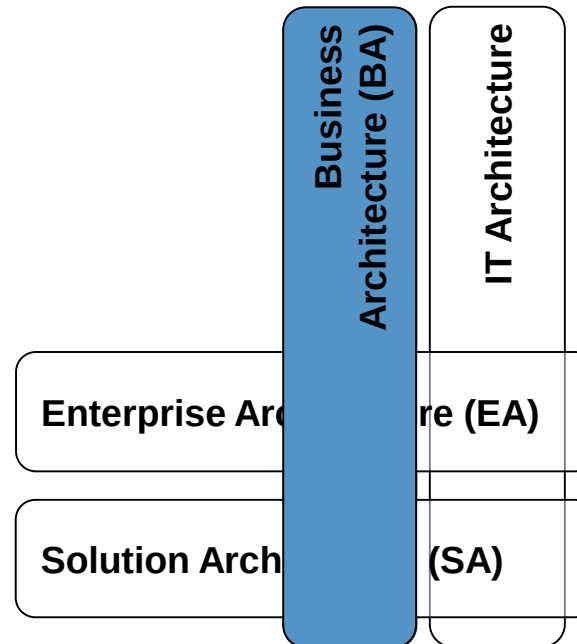
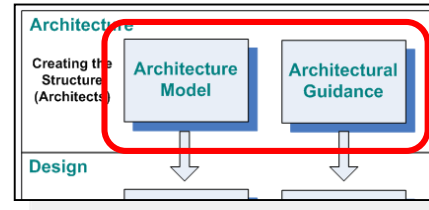


Architect supports two distinct dimensions (2 of 5)



Embracing both models and guidance:

- Business Architects “do” BA
 - *At both EA and SA levels.*

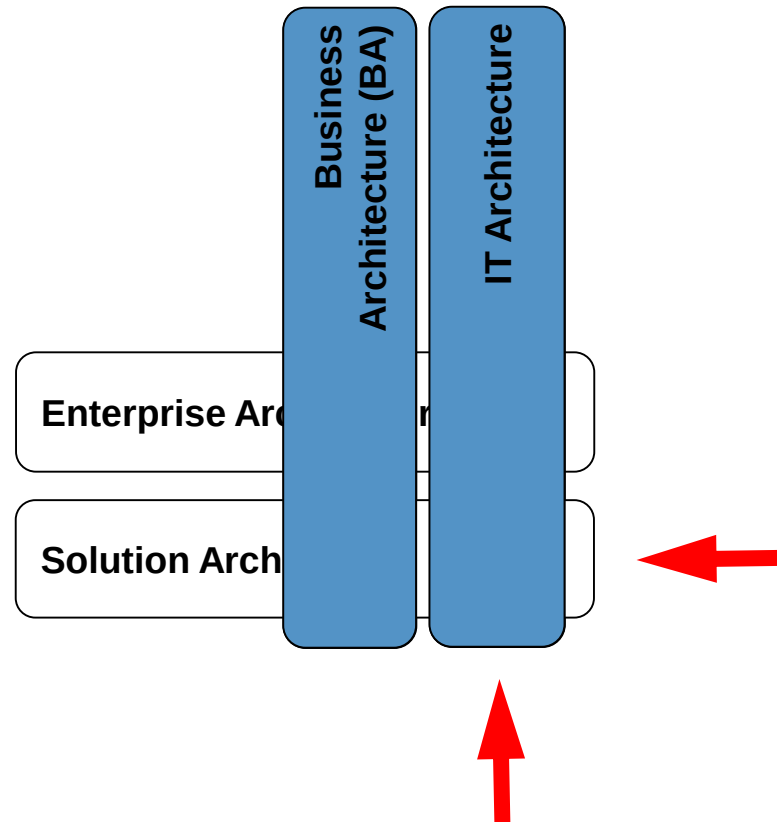
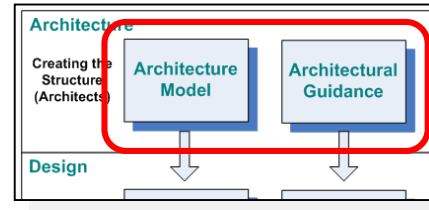


Architect supports two distinct dimensions (3 of 5)



Embracing both models and guidance:

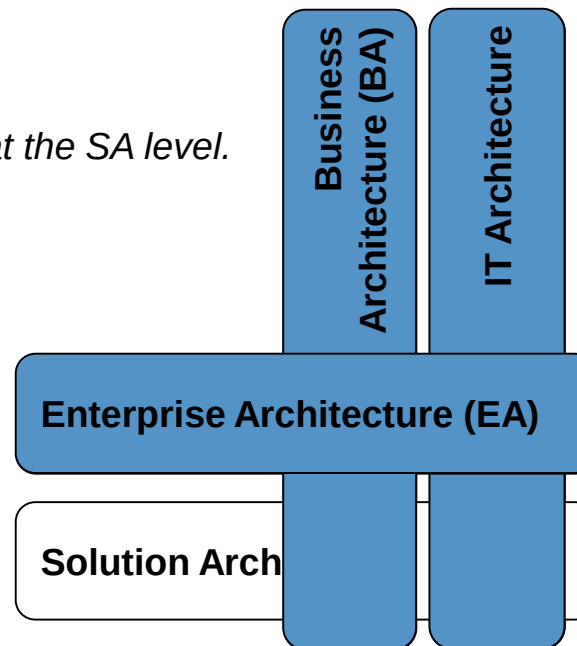
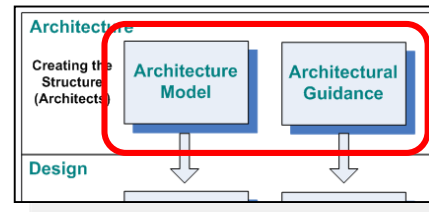
- Business Architects “do” BA
 - *At both EA and SA levels.*
- IT Architects “do” ITA
 - *Mostly at the SA level.*





Embracing both models and guidance:

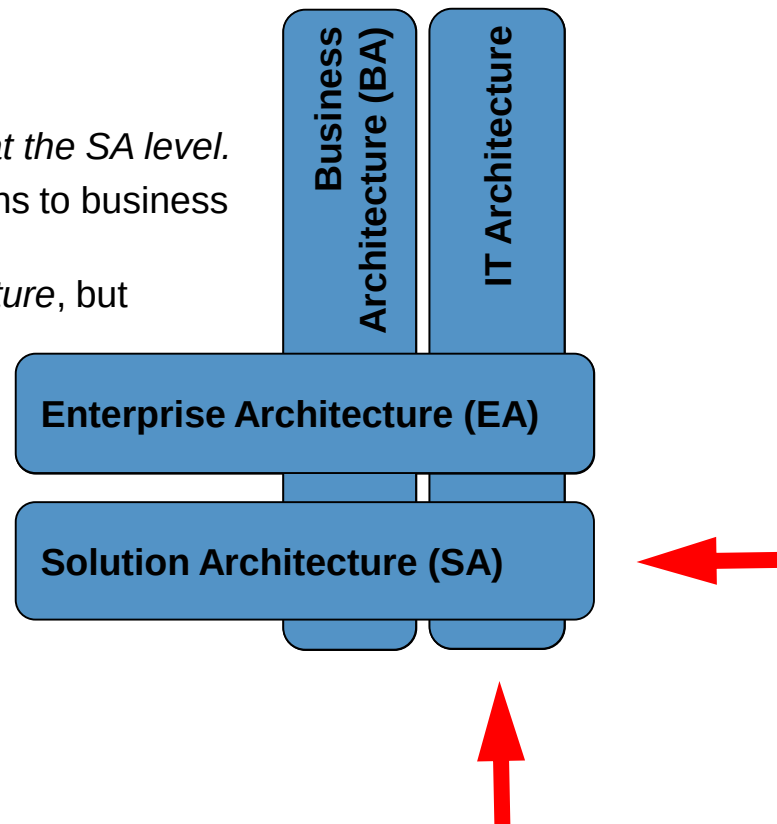
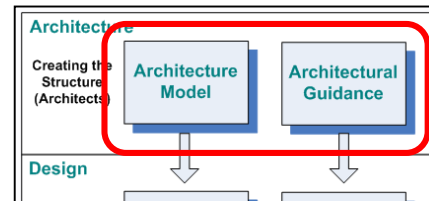
- Business Architects “do” BA
 - *At both EA and SA levels.*
- IT Architects “do” ITA
 - *Mostly at the SA level.*
- Enterprise Architects “do” EA
 - *Across both BA and ITA.*
 - *They often own guidance at the SA level.*





Embracing both models and guidance:

- Business Architects “do” BA
 - *At both EA and SA levels.*
- IT Architects “do” ITA
 - *Mostly at the SA level.*
- Enterprise Architects “do” EA
 - *Across both BA and ITA.*
 - *They often own guidance at the SA level.*
- Solution Architect create solutions to business problems – SA
 - *Typically within IT Architecture, but*
 - *Can create BAs too*





Time for a question



Which of the following choices is the better description of an Architect?

- A. A practitioner, a results-oriented person, a good communicator, a leader
- B. A “top-level” software designer, a project manager, a technology expert, a product expert, a lone scientist

Module outline



Introduction and objectives

What is architecture?

IBM's Architects

► **Summary and references**



- The architecture of a system describes the structure and behavior of its parts.
- The parts of an IT system are its software and hardware elements, interacting with the IT system's human and IT system users.
- An IBM Architect defines the models and guidance needed to model IT or business systems to meet the business intent.
- When working on an IT system solution architecture, the IBM Architect designs IT solutions to client business problems.

Back Up Slides: architecture vs Design



It is also possible to conceive of architecture and design in a rather different, possibly confusing, way (1 of 2)



In this alternative framework:

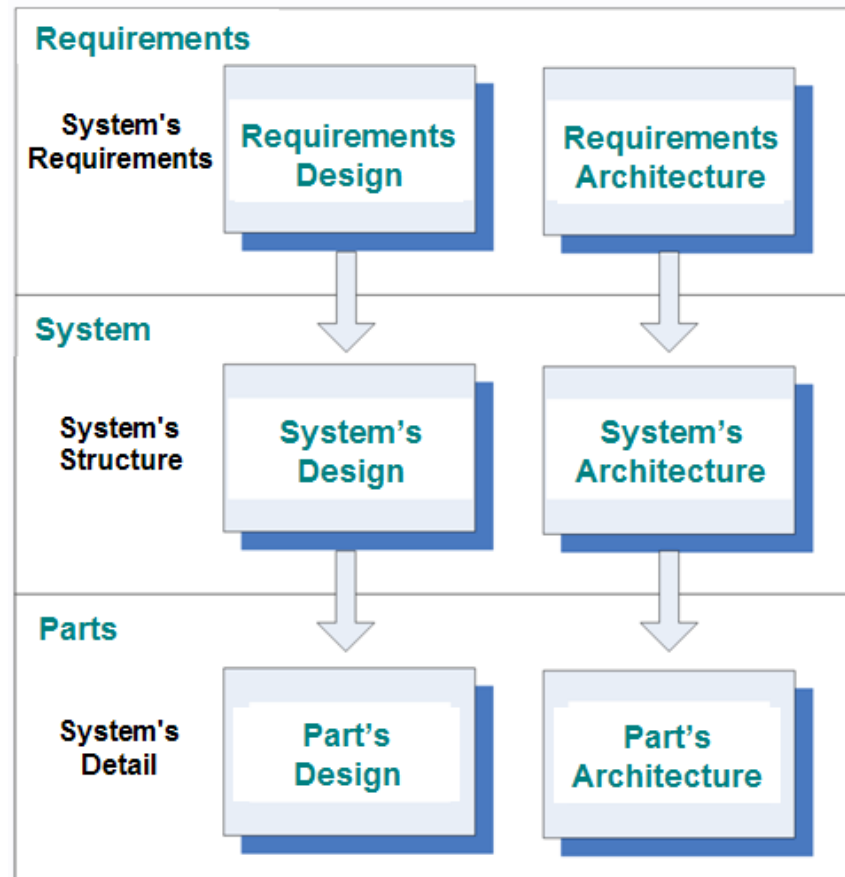
- Ensuring the system meets its specific requirements is called design.
- Ensuring it is designed in the right way is referred to as its architecture.

Ask yourself:

- What is the design of a bridge?
- What is the architecture of a bridge?

And:

- How many bridge architectures are there?
- How many bridges (that is, bridge designs) are there?



It is also possible to conceive of architecture and design in a rather different, possibly confusing, way (2 of 2)



In this alternative framework:

- Ensuring the system meets its specific requirements is called design.
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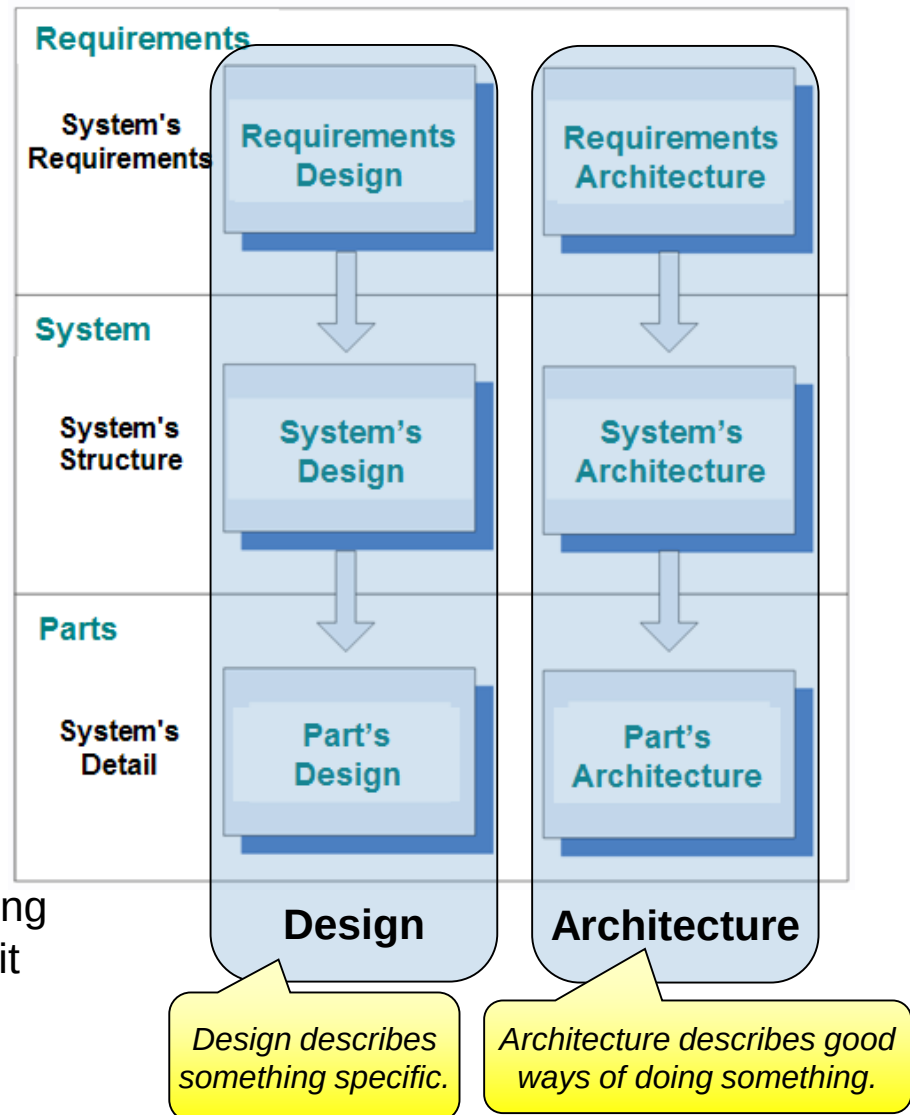
Ask yourself:

- What is the design of a bridge?
- What is the architecture of a bridge?

And:

- How many bridge architectures are there?
- How many bridges (that is, bridge designs) are there?

This terminology is common, particularly among our clients and in enterprise architecture, but it has not been adopted by IBM's Architect profession.





[COOK07] Cook, Denise, Cripps, Peter, Spaas, Philippe.

[An Introduction to the IBM Views and Viewpoints Framework for IT Systems](#), IBM,
December 2007

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